PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Discrimination capability of pre-test probability of stable coronary artery disease: a systematic review and meta-analyses suggesting how to improve validation procedures
AUTHORS	Mincarone, Pierpaolo; Bodini, Antonella; Tumolo, Maria Rosaria; Vozzi, Federico; Rocchiccioli, Silvia; Pelosi, Gualtiero; Caselli, Chiara; Sabina, Saverio; Leo, Carlo

VERSION 1 – REVIEW

REVIEWER	Wijns , William National University of Ireland Galway
REVIEW RETURNED	05-Feb-2021

GENERAL COMMENTS	The authors are proposing a careful pooled analysis of CAD pre-test probability models, showing significant heterogeneity. Suggestions are made in order to improve external validity of the models. Data are relevant to subjects with suspected chronic coronary syndromes. The analysis will be of interest to population scientists and statisticians, with limited impact on clinical practice. The main criticism refers to the decision to only consider the presence/absence of CAD on the basis of angiographic (anatomic) criteria. Incorporating the evidence of inducible ischemia (functional stenosis significance) would increase the relevance of the analysis. Page 5, lines 31+ : the diagnosis of “stable” CAD is actually a clinical diagnosis. It cannot be derived from functional testing or angiography. Please adjust the wording accordingly. Page 7, lines 12: as mentioned already, only considering anatomic criteria is a major limitation of this analysis. Page 23, lines 3-4: As mentioned, these findings only apply to Western populations.

REVIEWER	Dewey, Marc Charite Universitatsmedizin Berlin, Radiology
REVIEW RETURNED	18-Mar-2021

GENERAL COMMENTS	Review „Pre-test probability of stable coronary artery disease: a systematic review suggesting how to improve validation procedures“ This review addresses a pivotal question: how well do pretest probability predictions function and how can their methodological validation be improved. For this purpose the authors provide a comprehensive review of existing published tools.

	The current systematic review thus aimed to systematically assess and evaluate existing literature of pre-test probability models with an external validation as those are described having high level of quality for patient stratification based on a clinically driven pre-test probability assessment in patients with chronic coronary syndrome. The authors performed an extensive search in three databases (Global Health, Healthstar and MEDLINE) to identify eligible studies leading to 27 included studies on external validation of pre-test probability assessment in clinical studies with a stable chest pain cohort for diagnosis of coronary artery disease. Please find below my comments for further improving the quality of the manuscript. Abstract: - It is not clearly described which studies were evaluated for eligibility and inclusion. Thus, the reader may wonder what exactly this systematic review is about. Please add information under "Eligibility criteria" - The same applies for the methods section Introduction: - Second section is identical to the first section of the PROSPERO registry, I suggest rephrasing here in order to avoid duplication or self-plagiarism issues or actually referring to the PROSPERO entry section which was copied to the manuscript. - Page 6, line 17/18: Do you mean "8.92 million deaths"? - Page 7, last section: This sentence does not read well – comma fault? Methods: - Study inclusion criteria are not clearly explained (see also comment under Abstract). This makes it hard to fully understand the research question and the reports that are suggested to be studied. - Please elaborate further on the consensus reading - Was there an independent adjudicator in case of disagreement between the reviewers? - Why did you choose those three databases? Can you explain further and also comment on potential overlap between "Ovid HealthSTAR" (this refers to data from the National Library of Medicine's (NLM) MEDLINE) and "MEDLINE" (is this PubMed via MEDLINE?) – Please give more details - Also search strategy for each database should be outlined in - Were filters within the databases used? (e.g. see page #20) - Who are the so called randomly assigned reviewers? Are those colleagues of yours, which department and level of expertise? Or are those two of the authors listed? Please clarify this from a methodological point of view also in regard to authorship/acknowledgement aspects. This also should be clearly stated in the methods section - There seems to be some discrepancies compared to the PROSPERO registration: For example wording of the target patient population was described as "Patients with chest pain and a suspect of Stable CAD", in the manuscript you are referring to ""; also measurement of effects is described as "(…) by CCTA and ICA" vs. "CCTA or ICA"

	- Records identified should be shown for the databases separately. Results: - The authors stated that it was not possible to perform a meta-analysis approach due to high level of heterogeneity of studies included, two thoughts about this issue: - This may be the result of the eligibility and in-/exclusion criteria which were defined and could be discussed and explained as part of a given limitation. - A subanalysis could be possible to improve the overall generalizability and conclusion and I suggest the authors to check this option. - Flowchart: - Records screened (n=2685) – Records excluded (n=2442) = n=243, but only 83 full-texts were assessed Please explain discrepancies within the flowchart since as far as I understand there could be a calculation error. Discussion: - Page 52, line 12/13: Better discuss (treating or involved) physician(s) rather than only cardiologists – e.g. the patient's general practitioner or responsible physicians in the ER departments etc. have also a central role in evaluation of patients presenting with chest pain - Limitations of the present systematic review should be discussed as they appear to be missing in the current version.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Prof. William Wijns, National University of Ireland Galway

Comments to the Author:

- The authors are proposing a careful pooled analysis of CAD pre-test probability models, showing significant heterogeneity. Suggestions are made in order to improve external validity of the models. Data are relevant to subjects with suspected chronic coronary syndromes. The analysis will be of interest to population scientists and statisticians, with limited impact on clinical practice.The main criticism refers to the decision to only consider the presence/absence of CAD on the basis of angiographic (anatomic) criteria. Incorporating the evidence of inducible ischemia (functional stenosis significance) would increase the relevance of the analysis.

RESPONSE: First of all, many thanks for your valuable comments and suggestions.

Our review aims to have an impact on future studies on PTP development and validation as it mainly considers methodological issues. Additionally, we think that it could be of general interest to scientific societies as well as committed in developing guidelines and recommendations, and for clinicians who wish to be informed about the discussed methodological issues.

Regarding our choice of only considering the presence/absence of CAD on the basis of anatomic criteria, we are aware that incorporating the evidence of inducible ischemia would have increased the

value of our work. We reported our choice as a limitation (see at the end of the old "VALIDATED PTP MODELS: STRENGTHS AND WEAKNESSES" section, now *Discussion* section). This choice was based on the fact that, following the seminal work of Diamond and Forrester, several alternative PTP models have been developed in cohorts of patients referred for ICA or CCTA and a few of them were acknowledged also in Clinical Practice Guidelines. Indeed, due to its very high sensitivity and negative predictive values, CCTA substantially contributes to rule-out CAD. A previous systematic review on the same subject (Ting He et al. Clinics (Sao Paulo). 2017 Mar;72(3):188-196. doi: 10.6061/clinics/2017(03)10) found only one model developed to estimate the probability of functionally significant CAD (Caselli C et al, Can J Cardiol. 2015;31((6)):709–16. doi: 10.1016/j.cjca.2015.01.035). Caselli and colleagues introduced their work reporting that the available models had been tested only for their ability to predict the presence of anatomic CAD at invasive coronary angiography and not to predict functionally significant CAD, which is known to be associated with high prognostic risk.

- Page 5, lines 31+ : the diagnosis of "stable" CAD is actually a clinical diagnosis. It cannot be derived from functional testing or angiography. Please adjust the wording accordingly.

RESPONSE: We agree with you and have rephrased: "Stable CAD diagnosis is supported by non-invasive functional and/or anatomical testing,[2,3] and invasive coronary angiography (ICA)".

- Page 7, lines 12: as mentioned already, only considering anatomic criteria is a major limitation of this analysis.

RESPONSE: See our previous comment

- Page 23, lines 3-4: As mentioned, these findings only apply to Western populations.

RESPONSE: We have included this limitation in the discussion.

REMARK 1: Please, note that according to the suggestions of Reviewer 2 we added a meta-analysis. Then, the part of Results referring to Table 3 was modified accordingly. Most of the previous arguments are still included but rearranged due to the inclusion of the meta-analysis. For ease of rewriting, the whole section was cancelled and rewritten as you can see in the "Main Document - marked copy".

REMARK 2: Please, note that we have

- Modified the title of main sections (Results and Discussion)

- In the Data Extraction Strategy, we added “Correspondence with the authors of the included studies was initiated as necessary” which was not reported by a mistake in the previously submitted version.

Reviewer: 2

Dr. Marc Dewey, Charite Universitatsmedizin Berlin

Comments to the Author:

Review “Pre-test probability of stable coronary artery disease: a systematic review suggesting how to improve validation procedures“

This review addresses a pivotal question: how well do pre-test probability predictions function and how can their methodological validation be improved. For this purpose the authors provide a comprehensive review of existing published tools.

The current systematic review thus aimed to systematically assess and evaluate existing literature of pre-test probability models with an external validation as those are described having high level of quality for patient stratification based on a clinically driven pre-test probability assessment in patients with chronic coronary syndrome.

The authors performed an extensive search in three databases (Global Health, Healthstar and MEDLINE) to identify eligible studies leading to 27 included studies on external validation of pre-test probability assessment in clinical studies with a stable chest pain cohort for diagnosis of coronary artery disease.

Please find below my comments for further improving the quality of the manuscript.

RESPONSE: First of all, many thanks for your valuable comments and suggestions. Please find below our reply to your detailed indications.

- Abstract:
  - It is not clearly described which studies were evaluated for eligibility and inclusion. Thus, the reader may wonder what exactly this systematic review is about. Please add information under “Eligibility criteria”
  - The same applies for the methods section

RESPONSE: We have completely rewritten the Abstract trying to better clarify these points.

- Introduction:
  - Second section is identical to the first section of the PROSPERO registry, I suggest rephrasing here in order to avoid duplication or self-plagiarism issues or actually referring to the PROSPERO entry section which was copied to the manuscript.

RESPONSE: As suggested, we have rephrased the initial two paragraphs and updated the data on prevalence and mortality from the recently released study on Global Burden of Cardiovascular Diseases and Risk Factors.

- Page 6, line 17/18: Do you mean “8.92 million deaths”?

RESPONSE: Thank you, we have corrected.

- Page 7, last section: This sentence does not read well – comma fault?

RESPONSE: Thank you, we have fixed it.

- Methods:
  - Study inclusion criteria are not clearly explained (see also comment under Abstract). This makes it hard to fully understand the research question and the reports that are suggested to be studied.

RESPONSE: We have specified inclusion criteria in the section “Study inclusion and exclusion criteria”. In particular, we reported indications on Population, Intervention and Outcome.

- Please elaborate further on the consensus reading - Was there an independent adjudicator in case of disagreement between the reviewers?

RESPONSE: We have now specified that any disagreement was resolved by consensus among all the authors. We have also made clearer the whole procedure.

- Why did you choose those three databases? Can you explain further and also comment on potential overlap between “Ovid HealthSTAR” (this refers to data from the National Library of Medicine’s (NLM) MEDLINE) and “MEDLINE” (is this PubMed via MEDLINE?) – Please give more details

RESPONSE: We included GlobalHealth for the presence of the mapped topic “community and public health (including chronic diseases)”. We are aware of the potential overlap among Ovid HealthSTAR and MEDLINE but we just wanted to be as inclusive as possible.

- Also search strategy for each database should be outlined in

RESPONSE: Please note that the same search strategy reported as Additional File 1 has been applied to all the three databases at the same time. Consequently, the reported number of papers for each single line are collectively generated by the tool for the three sources. We have now specified this in the section “Searches” and in the Additional File 1.

- Were filters within the databases used? (e.g. see page #20)

RESPONSE: We have not used any filter within the databases. We have now specified this in the section “Searches”.

- Who are the so called randomly assigned reviewers? Are those colleagues of yours, which department and level of expertise? Or are those two of the authors listed? Please clarify this from a methodological point of view also in regard to authorship/acknowledgement aspects. This also should be clearly stated in the methods section

RESPONSE: We have rephrased the paragraph “Data extraction strategy” to specify that the randomly assigned reviewers are selected among the authors of the review. We have updated the role of the authors in the section “Authors' contribution”.

- There seems to be some discrepancies compared to the PROSPERO registration: For example wording of the target patient population was described as “Patients with chest pain and a suspect of Stable CAD”, in the manuscript you are referring to “”; also measurement of effects is described as “(…) by CCTA and ICA” vs. “CCTA or ICA”

RESPONSE: Many thanks for this detailed analysis which has allowed us two remove inconsistencies. Please see the changes in the section “Study inclusion and exclusion criteria” for patient population. We also amended some sections of PROSPERO, namely:

- INTERVENTION
    - OLD VERSION: Pre imaging/pre exercise test predictive model/score of CAD
    - NEW VERSION: pre-test models/scores of stable obstructive CAD adopted to address to guide clinician-patient decision-making for prevention and management of coronary artery disease
- MAIN OUTCOME(S)
    - OLD VERION: Probability of CAD
    - NEW VERSION: Discrimination capability of the pre-test models/scores vs. stable obstructive CAD defined by Coronary computed tomography angiography (CCTA) or Invasive coronary angiography (ICA)
- MEASURE OF THE EFFECT
    - OLD VERSION: effects measured by CCTA and ICA
    - NEW VERSION: Area under the Curve (AUC); sensitivity and specificity when available

Please note that for submissions from outside the UK (as in our case), the PROSPERO editorial team claims up to three months to process requests of changes.

- Records identified should be shown for the databases separately.

RESPONSE: We have tried to process your correct request. We know that the 2020 PRISMA Statement (BMJ 2021; 372 published on 29 March 2021) recommends to specify the number of records yielded per database. Unfortunately, the OVID search tool we used did not allow us to limit the search at a specific date (we can only limit it up to the end of a specific year) so we cannot run it again with a limit on 22nd April 2020. As you can see from the Supplementary Material - Additional File 1, the search was launched at the same time on all the considered databases without taking note of the results of each single source. Consequently, we are sorry but we can no longer retrieve the requested information.

- Results:
  - The authors stated that it was not possible to perform a meta-analysis approach due to high level of heterogeneity of studies included, two thoughts about this issue:
  - This may be the result of the eligibility and in-/exclusion criteria which were defined and could be discussed and explained as part of a given limitation.
  - A subanalysis could be possible to improve the overall generalizability and conclusion and I suggest the authors to check this option.

RESPONSE: Thank you for encouraging us to undertake this analysis. We carried-out a separate meta-analysis for the each PTP model, as a natural choice as different models have different predictors and weights. The "treatment effect" we considered was the AUC and we discussed our choice by adding a few words and references. In most of the cases, the analysis simply provided a summary AUC value with a poor reliability due to both the low amount of data for meta-analyses and the very high heterogeneity we obtained, preventing any meaningful interpretation.

Based on this updated approach, we focused on AUC values, modified Table 3 accordingly and added 3 figures. Few missing data were also filled in. A few comments on validations of AUC thresholds are now in the Discussion as a text. Please, note that the part of Results referring to Table 3 was modified accordingly. Most of the previous arguments are still included but rearranged due to the inclusion of the meta-analysis. For ease of rewriting, the whole section was cancelled and rewritten as you can see in the "Main Document - marked copy".

- Flowchart:
  - Records screened (n=2685) – Records excluded (n=2442) = n=243, but only 83 full-texts were assessed. Please explain discrepancies within the flowchart since as far as I understand there could be a calculation error.

RESPONSE: Thanks for having found it. It was a typo: Records excluded are 2,602

- Discussion:
  - Page 52, line 12/13: Better discuss (treating or involved) physician(s) rather than only cardiologists – e.g. the patient's general practitioner or responsible physicians in the ER departments etc. have also a central role in evaluation of patients presenting with chest pain

RESPONSE: Our intention was not specifically focused on what different health professionals are interested in evaluating but on highlighting the need for consistency between the diagnostic questions and investigation tools. Based on your comment we have decided to rephrase the sentence.

- Limitations of the present systematic review should be discussed as they appear to be missing in the current version.

RESPONSE: As suggested, we have added strengths and limitations of our work at the end of the Discussion section.

REMARK: Please note that we have

- Modified the title of main sections (Results and Discussion)
- In the Data Extraction Strategy, we added "Correspondence with the authors of the included studies was initiated as necessary" which was not reported by mistake in the previously submitted version.